

A Comparative Study of Anthropometric Characteristics Among Track And Field Athletes

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ABSTRACT

This study aims to explore the anthropometric characteristics of athletes participating in various track and field events, examining the role these traits play in athletic performance. The data for the present study have been collected on 103 male athletes from the different athletics events ranging from 15 to 22 years of age, who participated in Athletics trials held at NSNIS, Patiala. Anthropometric measurement has been taken on each subject by following the techniques of Weiner and Lourie (1969). Key Anthropometric variables such as height, weight, arm span, and several others were measured, and an analysis of variance (ANOVA) was conducted. The findings reveal statistically significant differences in several anthropometric variables among the athlete groups. Shot Putters had the highest mean weight and BMI, while runners exhibited the lowest in these categories. The results highlight the importance of specific anthropometric characteristics in different sports and suggest that targeted training regimens based on these traits could enhance the performance.

Keywords : Anthropometric characteristics, Physique, Body Mass Index (BMI), Waist-Hip Ratio (WHR).

INTRODUCTION

Performance in sports depends upon specific factors like size, shape, and proportion of an individual. The anthropometric profile of track and field athletes can vary depending on the particular event (Mooses et al., 2013). However, some general characteristics are commonly found in successful athletes (Sidhu, 2009). Sprinters, for example, tend to be shorter and more

muscular, with a higher percentage of fast-twitch muscle fibers. They typically have a mesomorphic body type, with broad shoulders and a narrow waist. Middle-distance runners are often leaner and have smaller body frames than sprinters (Khan et al., 2016). They have a more ectomorphic body type, long limbs and a slender build. Endurance runners, such as marathoners, also have a lean

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physique, but their body composition may vary depending on the specific event and training regimen. Jumpers and throwers tend to be taller and have a larger body frame than runners. They often have a higher percentage of slow-twitch muscle fibres and a more mesomorphic body type, with broad shoulders and a robust build.

Overall, the anthropometric profile of track and field athletes is highly specialized and optimized for their specific event. To perform at the highest level, successful athletes must have a combination of physical traits, such as power, speed, endurance, technical skill, and mental toughness (Mészáros et al., 2000). Anthropometric profiling is a widely used method for analyzing athletes' physical characteristics and developing training programs specific to their sport. In track and field, anthropometric profiling is a valuable tool for identifying key performance indicators and optimizing training and competition strategies.

Studies have shown that specific physical characteristics, such as body mass index (BMI), body fat percentage, and limb length ratios, are associated with success in different track and field events (Manning & Pickup, 1998; Mooses et al., 2013). In addition to body type and muscle fiber composition, other anthropometric factors that can impact athletic performance include height, weight, limb length, body fat percentage,

and bone density. In particular, a high bone density can support the muscles more significantly and reduce the risk of stress fractures, which are common in distance runners. There are also gender differences in the anthropometric profile of track and field athletes. Women tend to have a higher body fat percentage and a smaller body frame than men, which can impact their athletic performance. However, women also have a more excellent range of motion in their joints, which can be an advantage in events that require flexibility, such as gymnastics. Thus, the present study aims to explore the anthropometric characteristics of athletes participating in various track and field events, examining the role these traits play in athletic performance.

MATERIAL AND METHODS

The study was conducted on 103 male athletes age range from 15 to 22 years attended Athletics trial on February 2023 held at NSNIS, Patiala. The data have been collected using standardized techniques and instruments. The measurements include height, weight, sitting height, arm span, circumference of waist-hip, acro - radiale length, radiale - stylium length, mid stylium - dactylium length, height trochanterion, height tibiale, bi-iliac diameter, bi-acromial diameter, right & left foot length. A Digital BMI machine (Seca-284), Harpenden sitting height table, anthropometric rod (GPM), and

measuring tape (Hoeschstmass, West Germany)) were used for the measurements.

The statistical analysis has been done to calculate the mean, standard deviation, student's t-test and F value by using SPSS version 20. One-way analysis of variance (ANOVA) was computed to compare the events on the selected anthropometric variables, then the Scheffe post-hoc test was applied for the statistical significance. Confidence Interval level for statistical significance

was fixed at 95% ($p < 0.05$).

RESULTS

The table 1 presents athletes participating in various track and field events, totaling 103 athletes. It categorizes the athletes into seven distinct events: Discus Throwers 12 (11.7%), Hammer Throwers 6 (5.8%), Javelin Throwers 13 (12.6%), Long Jumpers 20 (19.4%), Shot Putters 10 (9.7%), Sprinters 31 (30.1%), and Triple Jumpers 11 (10.7%).

Table 1: Analysis of the variance of anthropometric variables of the Athletes

VARIABLES	Discus Throwers (N = 12) 1		Hammer Throwers (N = 6) 2		Javelin Throwers (N = 13) 3		Shot putters (N = 10) 4		Long Jumpers (N = 20) 5		Triple Jumpers (N = 11) 6		Sprinters (N = 31) 7		F Ratio
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	
Height (cm)	182.93 ^{5,7}	6.03	177.05	4.65	176.38	5.22	179.15	5.36	175.11	5.67	174.26	7.53	173.21	5.52	4.89**
Weight (Kg)	88.98 ^{5,6,7}	14.52	84.28 ^{5,6,7}	10.3	76.06 ⁴	8.53	100.66 ^{3,5,6,7}	13.45	66.1	6.81	61.4	7.74	65.75	9.45	26.22**
Sitting Height (cm)	92.87 ^{5,6,7}	3.77	89.97	2.33	90.49	2.9	91.89	2.48	88.86	2.71	88.3	3.4	89	2.73	4.40**
Arm span (cm)	194.26 ⁷	8.11	184.43	4.78	186	5.12	190.82	9.41	183.71	7.98	182.62	11.27	180.6	8.24	5.14**
Body Mass Index	26.52 ^{4,5,6,7}	3.58	26.82 ^{5,6,7}	2.3	24.41 ^{4,6}	2.13	31.34 ^{1,3,5,6,7}	3.78	21.54	1.82	20.16	1.62	21.84	2.34	28.69**
Waist-hip Ratio	0.83 ⁴	0.06	0.87	0.03	0.83 ⁴	0.03	0.90 ^{1,3,5,6,7}	0.04	0.82	0.03	0.82	0.02	0.83	0.03	7.72**
Sitting Height Ratio	50.77	1.28	50.82	0.58	51.31	1.11	51.3	0.78	50.76	1.07	50.7	1.33	51.39	0.96	1.42
Acro-radiale	36.51 ^{6,7}	1.7	33.05	1.86	33.88	1.54	35.41 ⁷	2.77	34.04	1.76	33.31	2.6	32.95	1.93	5.83**
Radiale-stylion	27.92	1.77	27.13	3.01	26.19	1.15	27.3	2.11	26.21	1.48	25.83	2.23	25.83	1.45	2.98*
Midstylion-Dactylion	19.48	1.06	19.18	0.86	19.52	1.11	19.22	1.3	19.32	1.13	19.29	1.82	19.01	1.23	0.37
Height Trochanterion	96.86 ⁷	4.49	93.85	3.88	94.23	3.91	94.3	3.56	94.12	3.09	94.13	5.45	91.84	3.57	2.63*
Height Tibiale	49.74	2.19	48.72	2.49	48.4	1.73	48.83	2.89	47.86	2.25	47.23	3.87	46.63	2.63	2.75*
Bi-iliac Diameter	29.74 ^{5,6,7}	1.97	29.77 ^{5,6,7}	1.65	28.04 ⁴	1.75	31.51 ^{3,5,6,7}	1.94	26.61	1.38	26.1	1.86	26.84	1.65	16.94**
Bi-acromial Diameter	42.5 ⁷	2.72	41.37	0.51	41.67	2.11	44.02 ^{5,6,7}	1.68	40.18	1.89	39.59	2.31	39.95	1.86	8.03**
Right Foot	27.62 ^{5,7}	0.85	26.53	1.22	26.43	0.97	27.65 ^{5,7}	0.95	26.02	1.26	26.25	1.51	25.75	1.15	6.24**
Left Foot	27.8 ^{5,7}	0.93	26.72	0.93	26.45	0.84	27.6 ⁷	0.87	26.06	1.29	26.23	1.48	25.79	1.11	7.03**

*The mean difference is significant at the 0.05 level (2.19), N=103, df = 101, 06

**The mean difference is significant at the 0.01 level (2.99) N=103, df = 101, 06

The analysis of variance (ANOVA) of various anthropometric variables among athletes participating in seven different sporting events is shown in table 1. The table includes key measurements

such as height, weight, sitting height, arm span, body mass index (BMI), waist-hip ratio (WHR), and several other specific anthropometric variables.

The F value displays the statistical

significance of the differences among the groups. For instance, the F value for weight is notably high at 26.22 ($p < 0.01$), suggesting a statistically significant variance in weight across the groups. Shot putters exhibited the highest mean weight at 100.66 kg, while Sprinters had the lowest at 65.75 kg. Similarly, height also displayed significant differences among the athlete categories, evidenced by an F value of 4.89 ($p < 0.01$). Discus throwers were the tallest on average with a height of 182.9 cm, contrasting with Sprinters, who were the shortest with an average height of 173.21 cm.

Derived variables like BMI and WHR also presented statistically significant differences among the groups. The F value for BMI stood at a remarkable 28.69 ($p < 0.01$), highlighting the substantial variations in this metric. Shot Putters had the highest average BMI of 31.34 kg/m², whereas Triple Jumpers recorded the lowest at 20.16. WHR differences were likewise significant, as indicated by an F value of 7.72 ($p < 0.01$). Shot putters had the highest WHR at 0.90, while Long and Triple Jumpers had the lowest at 0.82.

Furthermore, the table reports other specific anthropometric measurements, such as arm span and various diameters, each with its own set of statistically significant differences as indicated by the F value. Arm span showed a significant F value of 5.14 ($p < 0.01$), with Discus

Throwers having the longest mean arm span of 194.26 cm, and Sprinters the shortest at 180.6 cm. Sitting Height and Midstylium dactylium had shown non-significant among throwers, jumper and sprinters. Acro-radiale (upper arm) length and radiale-stylium had significant ANOVA F values at 0.05 and 0.01 level. Maximum values were observed in discus thrower followed by shot putters, long jumpers, javelin throwers, triple jumpers, hammer thrower and least in sprinters. Height trochanterion (Leg length) and height tibiale (lower leg length) were observed significant F value at 0.05 level among throwers, jumpers and sprinters. Maximum value of lower limbs were examined in discus thrower and minimum in sprinters. Bi-acromiale and bi-iliac diameters were recorded significant F values at 0.01 level. Maximum diameters were observed in shot putters and least in sprinters. Foot size variables also reflected significant differences, with F value for the right and left foot at 6.24 ($p < 0.01$) and 7.03 ($p < 0.01$), respectively.

Overall, Table 1 provides a comprehensive overview of the anthropometric characteristics of athletes across different sporting events. The F value indicate where these differences are statistically significant, providing a nuanced understanding that could have implications for athletic performance and sport-specific training regimens.

DISCUSSION

The age range of 15 to 22 years in this study is significant because this is a critical period for physical development and maturation. Understanding the anthropometric profile of male athletes within this age range can be particularly useful for identifying potential talents and selecting athletes for specific events. The measurements taken in this study can provide insights into an athlete's body composition and skeletal structure, which are critical factors in determining athletic performance. For example, the biacromial diameter, the distance between the two shoulders, is important in throwing events such as shot put and discus throw. A wider bi-acromial diameter can provide an advantage in these events, generating greater force during the throwing motion.

Similarly, height, arm span, and leg length are important factors in jumping events such as high and long jumps. Taller athletes with longer limbs may be advantageous in these events as they can generate more power and cover a greater distance.

In addition to identifying physical strengths and weaknesses, understanding an athlete's anthropometric profile can also help with injury prevention. For example, if an athlete has a lower bone density, they may be at a higher risk for stress fractures, which can impact their ability to train and compete. By

understanding these risks, coaches, and trainers can adjust an athlete's training program to minimize the risk of injury.

Gharbi et al. (2020) found that sprinters had significantly lower body fat percentages and higher muscle mass compared to distance runners. Another study by Malina et al. (2017) found that height, body mass, and BMI were positively associated with success in the high jump event. Knechtle et al. (2011) found that longer lower limb lengths were associated with better performance in the triple jump event. Overall, anthropometric profiling can provide valuable information for coaches and sports scientists in track and field to optimize training programs and improve performance in competition.

CONCLUSION

The present study aimed to compare the anthropometric variables of seven different groups of athletes. The analysis showed significant differences in most of the variables among the groups. In terms of height, discus throwers had the highest mean (182.93 cm), followed by shot-putters (179.15 cm), hammer throwers (177.05 cm) and javelin throwers (176.38 cm). Sprinters, Long and Triple jumpers had the lowest mean of height 173.21cm, 175.11 cm and 174.26 cm, respectively. The F value indicated a significant difference among the groups. This finding suggests that height may be

crucial in determining an athlete's suitability for certain events.

Regarding weight, shot putters had the highest mean (100.66 kg), while triple jumpers had the lowest mean (61.4 kg). The F value showed a significant difference among the groups. This finding implies that body mass determines an athlete's performance in specific events.

In terms of sitting height, discus throwers had the highest mean (92.87 cm), followed by shot-putter (91.89), javelin throwers (90.49 cm) and hammer throwers (89.97 cm). Long and triple jumpers had the lowest means (88.86 cm and 88.3 cm, respectively). The F value showed a significant difference among the groups. This finding suggests that the sitting height may be related to the length of the torso, which may be an important factor in determining an athlete's performance in throwing events.

Regarding arm span, discus throwers had the highest mean (194.26 cm), followed by shot-putter (190.82), javelin throwers (186 cm) and hammer throwers (184.43 cm), triple jumpers and Sprinter had the lowest means (182.62 and 180.6 cm, respectively). The F value showed a significant difference among the groups. This finding implies that arm span may be crucial in determining an athlete's suitability for certain events requiring longer reaches, such as throwing and jumping.

Regarding BMI, shot putters had the highest mean (31.34 kg/m²), while long jumpers and triple jumpers had the lowest mean (21.54 and 20.16 kg/m², respectively). The F value showed a significant difference among the groups. This finding suggests that BMI may be related to an athlete's suitability for certain events, such as throwing events, where greater body mass may be advantageous.

Regarding the waist-to-hip ratio (WHR), shot-putters had the highest mean (0.875), while long jumpers and triple jumpers had the lowest mean (0.82). The F value showed a significant difference among the groups.

In conclusion, this study's findings indicate that height, weight, sitting height, arm span, BMI, and WHR are essential anthropometric variables that may influence an athlete's performance in specific events. Further research can explore other anthropometric variables that may be important in determining an athlete's suitability for specific events. The findings of this study suggest that anthropometric variables are essential in determining an athlete's suitability for specific events. Coaches and trainers can use these findings to select athletes with the right physical attributes for particular events. Athletes can also use this information to determine which events suit them best and adjust their training accordingly.

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